

Internet of Things

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IR Sensor

- In a world increasingly driven by technology, the Infrared Sensor, or IR Sensor, emerges as a pivotal component in numerous applications.
- It unravels their fundamental nature and their critical role in modern technological advancements.
- IR Sensors have become indispensable to our technological landscape, from simplifying everyday tasks to enabling complex machinery.
- As we delve into their intricacies, we discover how they function and shape the interaction between technology and our daily lives.

What is an IR Sensor?

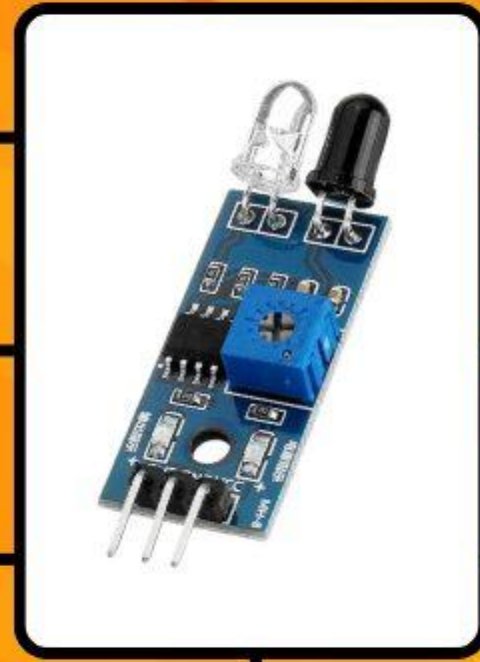
An Infrared Sensor, commonly known as an IR Sensor, is a pivotal component in modern technology, particularly sensor systems. Fundamentally, it's a device that uses infrared light to detect and measure various aspects of its environment. By emitting or detecting infrared radiation, these sensors can perceive the physical world in ways beyond human capabilities.

WHAT IS IR SENSOR?

Proximity Sensing Technology

Motion Detection Capabilities

Measures Heat Emissions



Detects Infrared Radiation

Uses Infrared Light

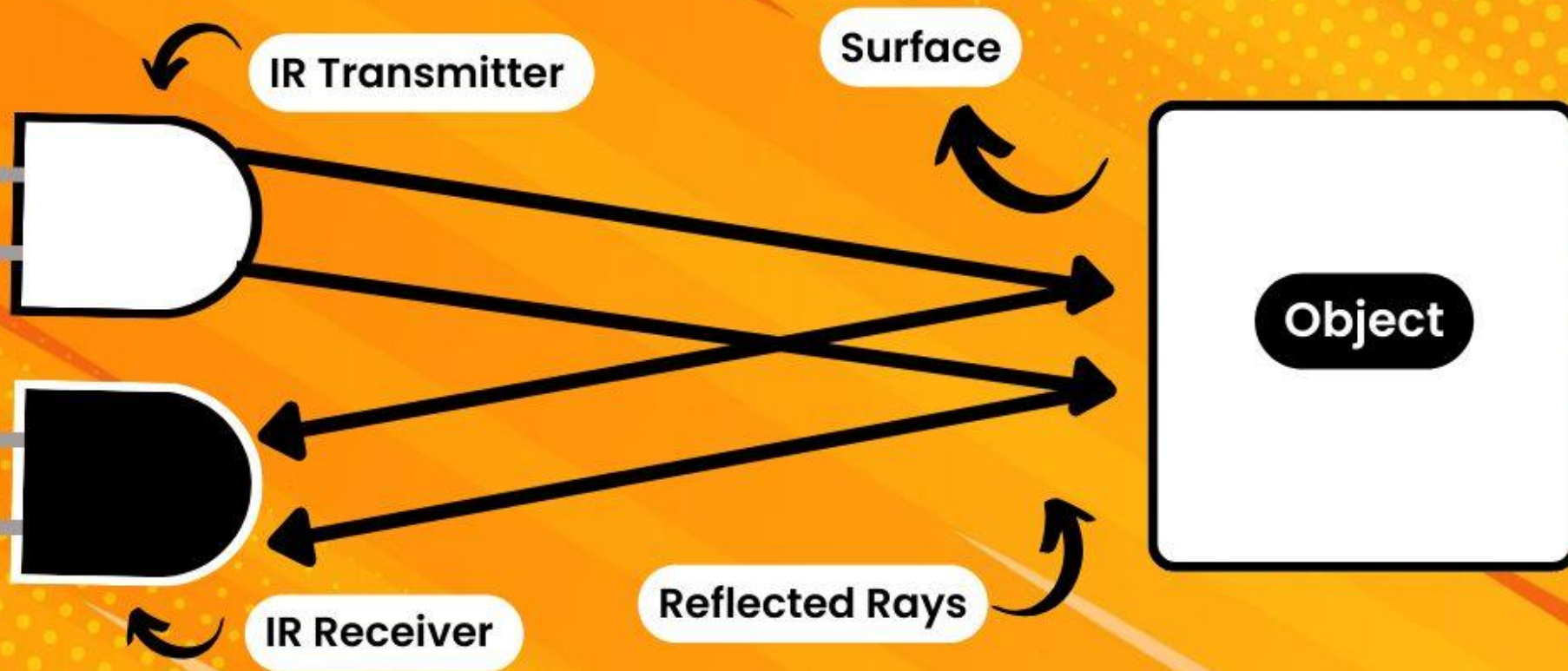
Obstacle Avoidance Systems

Used in Remote Controls

Working Principle of IR Sensors

- The IR LED and the Photodiode are at the heart of an IR Sensor's functionality.
- The IR LED emits infrared light, which gets reflected upon hitting an object and is detected by the Photodiode.
- This simple yet effective mechanism enables an IR Sensor to perform its task.

WORKING PRINCIPLE OF IR SENSORS



Types of IR Sensors and Their Uses

There are two main types:

- Active and Passive IR Sensors.
- **Active IR Sensors** are the more interactive ones. They emit infrared light and detect its reflection, making them perfect for applications requiring direct measurement.
- **Passive IR Sensors** are often used in security systems. They work by detecting the infrared radiation emitted by warm objects like human bodies.

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- Active IR Sensors create and detect signals. In contrast, Passive IR Sensors quietly observe and pick up infrared energy. This difference leads to diverse applications.
- Active IR Sensors excel in targeted detection, while Passive IR Sensors are preferred for surveillance and motion detection.

Active IR Sensors

- Active IR sensors are the cornerstone of precision in the sensor technology landscape.
- These sensors operate by emitting a beam of infrared light, which reflects off objects and returns to the sensor.
- This mechanism allows for exceptionally accurate object detection, making them indispensable in applications requiring high precision.

For instance, active IR sensors reliably detect and count products in automated manufacturing lines, ensuring efficiency and accuracy.

WHAT IS ACTIVE IR SENSOR?



Offers High Sensitivity

Detects Infrared Radiation

Active Sensing Technology

Emits IR Signals

Requires Power Source

Measures Distance Accurately

Detects Object Presence

Passive IR Sensors

- Passive IR Sensors, a marvel in sensor technology, stand out for their ability to detect natural infrared radiation emitted by objects.
- Unlike their active counterparts, these sensors do not emit any infrared light; instead, they are finely tuned to sense the subtle infrared energy naturally given off, especially by warm objects like the human body. This unique capability makes them incredibly useful in various applications.

For instance, in home security systems, passive IR sensors provide a reliable and non-intrusive way to detect potential intruders by sensing body heat.

WHAT IS PASSIVE IR SENSOR?



Motion Sensing Technology

Utilizes Passive Infrared

Common in Security Systems

Used in Lighting Control

Simple Installation Process

Functions in Total Darkness

Long Lifespan Components

Active vs Passive

ACTIVE VS PASSIVE IR SENSORS



Active IR Sensor

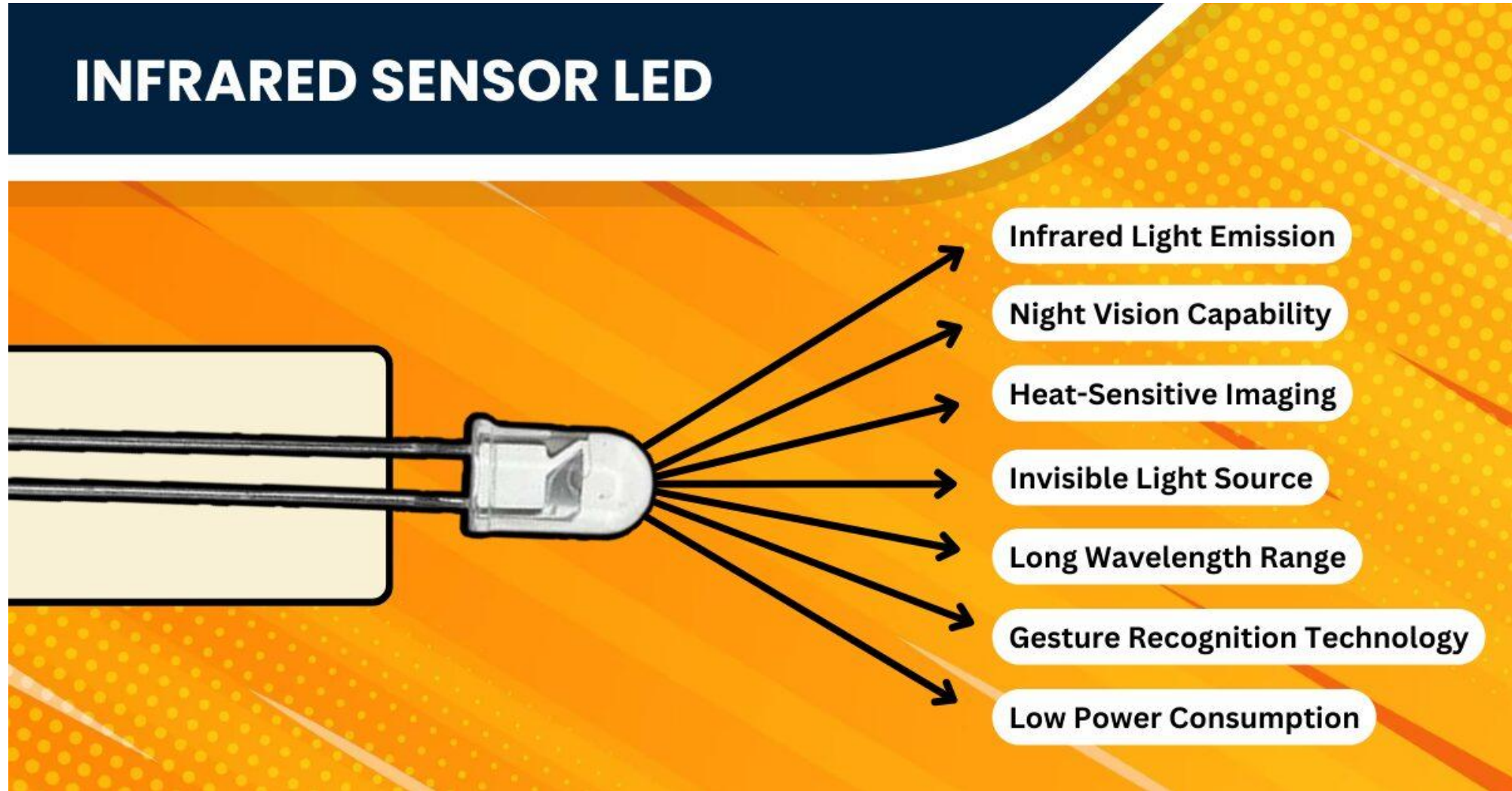
- ◆ Emits Infrared Light
- ◆ Detects Reflected Signals
- ◆ Shorter Detection Range
- ◆ Higher Energy Consumption
- ◆ Limited Outdoor Use
- ◆ Ideal for Controlled Environments



Passive IR Sensor

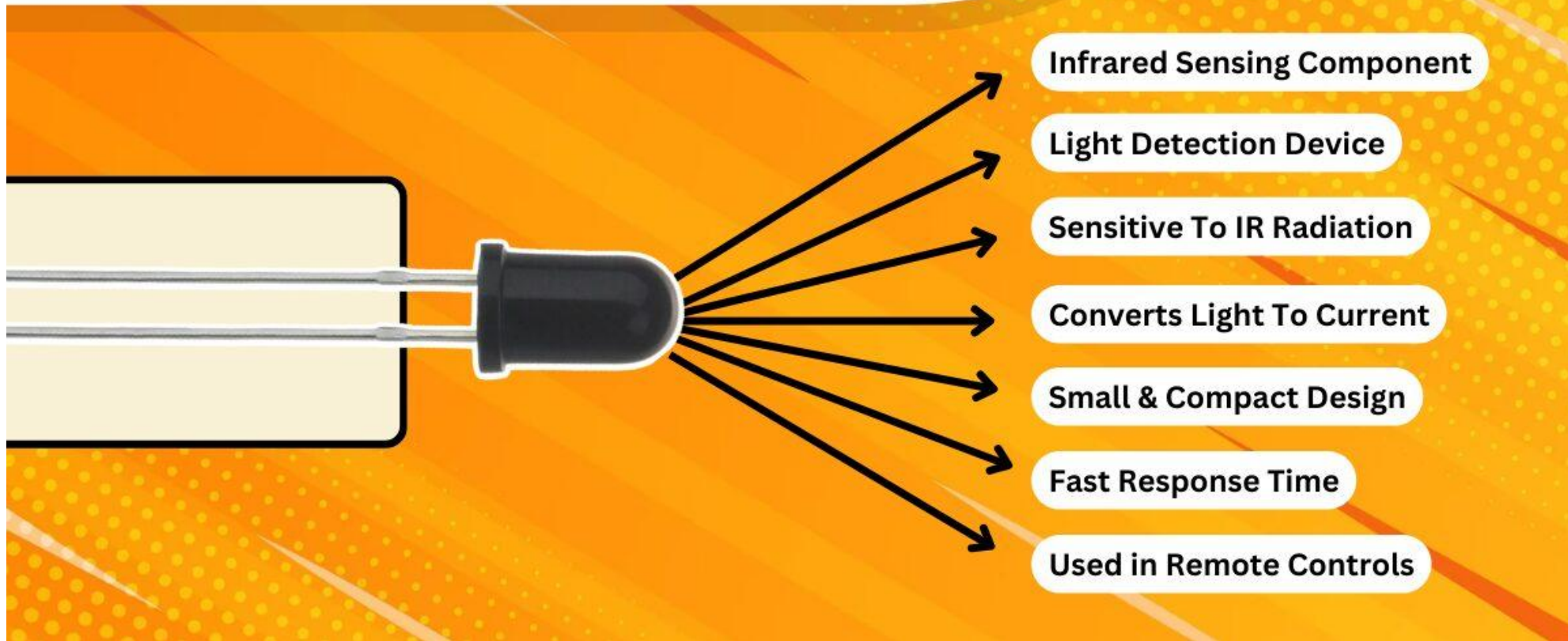
- ◆ No Emission of Light
- ◆ Detects Infrared Radiation
- ◆ Longer Detection Range
- ◆ Low Energy Consumption
- ◆ Effective in Outdoor Settings
- ◆ Adaptable to Various Conditions

IR LED: The Beacon of Infrared Light



Photodiode

PHOTODIODE OF IR SENSOR



DEVICES REQUIRED

- Arduino Board.
- Jumper Cables.
- A to B Cable.
- LED Module.
- IR Module.

Assignment

- 1. Interface with the IR Sensor and detect the obstacle.**
- 2. Add an RGB LED with the IR and turn it Red and Green based on the detection of an Obstacle.**

Please write the same in your Lab Record Books with the Circuit Diagram on the blank page, and upload your codes to Teams as .ino for each assignment. Please sign each code with your name and roll number in the comments.

Thanks